

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - V

Roll No. : S003	Name: Aryan Raj
Class : SYIT	Batch : –
Date of Assignment : 12-08-26	Date/Time of Submission : 12-08-26

Part – A Theory

5a https://www.w3schools.com/python/pandas/ref_df_filter.asp

5b https://www.w3schools.com/python/module_statistics.asp

5c <https://www.geeksforgeeks.org/python/creating-a-pandas-series-from-dictionary/>

5d <https://www.geeksforgeeks.org/pandas/boolean-indexing-in-pandas/>

PART – B Practical Questions:

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

Code:

```
import pandas as pd

data = {
    "name": ["Sally", "Mary", "John"],
    "age": [50, 40, 30],
    "qualified": [True, False, False]
}
df = pd.DataFrame(data)

print(df)
print("Aryan S003")
```

Output:

```
=====
      name  age  qualified
0  Sally   50      True
1  Mary   40      False
2  John   30      False
Aryan S003
|
```

5b. Write a python code to show statistical information on given data set.

Code:

```
practical 5.py - D:/Project File/pythan for ds/practical 5.py (3.14.6)
File Edit Format Run Options Window Help
import statistics

print("Mean:")
print(statistics.mean([1, 3, 5, 7, 9, 11, 13]))
print(statistics.mean([1, 3, 5, 7, 9, 11]))
print(statistics.mean([-11, 5.5, -3.4, 7.1, -9, 22]))

print("\nMedian:")
print(statistics.median([1, 3, 5, 7, 9, 11, 13]))
print(statistics.median([1, 3, 5, 7, 9, 11]))
print(statistics.median([-11, 5.5, -3.4, 7.1, -9, 22]))

print("\nMode:")
print(statistics.mode([1, 3, 3, 3, 5, 7, 7, 9, 11]))
print(statistics.mode([1, 1, 3, -5, 7, -9, 11]))
print(statistics.mode(['red', 'green', 'blue', 'red']))

print("\nStandard Deviation:")
print(statistics.stdev([1, 3, 5, 7, 9, 11]))
print(statistics.stdev([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.stdev([-11, 5.5, -3.4, 7.1]))
print(statistics.stdev([1, 30, 50, 100]))

print("\nVariance:")
print(statistics.variance([1, 3, 5, 7, 9, 11]))
print(statistics.variance([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.variance([-11, 5.5, -3.4, 7.1]))
print(statistics.variance([1, 30, 50, 100]))

print("Aryan S003")
|
```

Output:

```
Mean:
7
6
1.8666666666666667

Median:
7
6.0
1.05

Mode:
3
1
red

Standard Deviation:
3.7416573867739413
0.6925797186365383
8.414471660973927
41.67633221226007

Variance:
14
0.47966666666666663
70.80333333333333
1736.9166666666667
Aryan S003
```

5c. Write a python code to create pandas series from dictionaries.

Code:

```
import pandas as pd

dictionary = {'DS': 10, 'PDS': 20, 'AM': 30}

series = pd.Series(dictionary)

print(series)
print("Aryan S003")
|
```

Output:

```
DS      10
PDS     20
AM      30
dtype: int64
Aryan S003
|
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

Code:

```
import pandas as pd

dict = {'name':["aparna", "pankaj", "sudhir", "Geeku"],
        'degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'score':[90, 40, 80, 98]}

df = pd.DataFrame(dict, index = [True, False, True, False])

print(df)
print("Aryan S003")
```

Output:

```
=====
      name  degree  score
True  aparna    MBA    90
False  pankaj    BCA    40
True   sudhir  M.Tech    80
False   Geeku    MBA    98
Aryan S003
```